THREE OUTCOMES OF ONE RULE

Dividing two powers of the same base subtracts the exponents. Whether $m - n$ comes out positive, zero or negative decides what the result looks like.

QUOTIENT RULE

$$\frac{a^m}{a^n} = a^{m-n}$$

split by m compared with n

requires $a \neq 0$

+

CASE 1

$m > n$

A positive power

a^{m-n} with $m - n > 0$

E.G. $\frac{x^7}{x^2} = x^5$

1

CASE 2

$m = n$

The result is 1

$a^0 = 1$

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CASE 3

$m < n$

A negative exponent

a reciprocal; see negative exponents

E.G. $\frac{5^3}{5^5} = 5^{-2} = \frac{1}{25}$

TAKEAWAY

All three cases are natural consequences of subtracting exponents, and all three require $a \neq 0$ and identical bases.